Literature Review: Deep Learning for Seizure Detection and Prediction from Non-EEG Wearable Signals (2013–2026)

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Contents

1 Introduction

1.1 Problem Statement and Motivation

Epilepsy, a chronic neurological disorder affecting 60 million people wordlwide, with approximately one-third of patients remaining drug-resistant (Hussein et al., 2018). Sveinsson et al., 2020 found that 69% of Deaths due to a Generalized tonic-clonic seizure could have been prevented if not left unattended, highlighting the demand for reliable and timely prediction.

While reliable prediction is already possible through electroencephalography (EEG), its operational complexity render it unsuitable for ambulatory use. To address the need for ambulatory detection many are experimenting with utilizing autonomic and motion signals to detect and predict seizures.

However, translating and analysing these often noisy and complex time-series physiological signals has pushed traditional machine learning (ML) approaches to their limits. These approaches require manual engineering such as cleaning and extracting of the data, often failing to capture the intricate and non-linear complex temporal dynamic of these signals. It is this limitation that deep learning (DL) models are designed to overcome, with their ability of hierarchical and learned feature extraction uniquely addressing these limitations.

This review therefore synthesizes and critically evaluates recent advances in time-series deep learning—particularly convolutional neural networks (CNNs) and long short term memory (LSTMs)— applied to seizure detection and prediction using non-EEG wearable biosignals. Central to this analysis is the question of how architectural choices, training paradigms (e.g., transfer learning, personalization), and rigorous, prospective evaluation can converge to meet the stringent clinical requirements of high sensitivity and minimal false alarm rates (FAR) in real-world deployment.

subsectionResearch Aim and Questions How do different deep learning architectures and biosignal modalities excluding EEG compare in their ability to achieve a optimal trade-off between sensitivity and false alarm rate for ambulatory seizure monitoring?

2 Theoretical Foundations for Ambulatory Seizure Monitoring

2.1 The Ambulatory Monitoring Paradigm: Requirements and Challenges

Beniczky et al., 2021, Chen et al., 2022 found that translation of seizure detection from the controlled clinical environment to an ambulatory setting redefines approaches and performance requirements. In clinical environments maximum sensitivity is prioritized over FAR, when engineering for ambulatory use a balance of both metrics is needed. Failing to detect a seizure (low sensitivity) poses a safety risk, While excessive false alarms (high FAR) leads to desensitizing and "alarm fatigue" and the device being ignored or not used at all. Users reported that devices should achieve high sensitivity ($\geq$90%) with low false alarm rates — acceptable FARs were 1-2/week for patients with $\geq$1 seizure/week and 1-2/month for patients with $<$1 seizure/week (Sivathamboo et al., 2022). While more complex models can find more intricate patterns, they also introduce the "black-box" problem wherein the model lacks transparency AbuAlrob et al., 2025; Ghaderi, 2025. Therefore, the design should incorporate explanations for decisions to ensure user trust and regulatory compliance AbuAlrob et al., 2025; Miron et al., 2025.

2.2 Physiological Biosignals for Wearable Sensing

Wearable devices circumvent the impracticality of EEG by capturing peripheral biosignals that serve as proxies for central nervous system activity during seizures. These proxies will inherently relay incomplete and noisy signals so a combination of multiple sensors is required. The acceleration/attitude sensor excels at detecting muscle convulsions but is noisy in daily use, as it captures all movement. The electrodermal activity (EDA) sensor detects autonomic arousal by measuring the skin conductance associated with sweat gland activity.

2.3 Seizure Manifestations and Corresponding Biosignals

Wearable biosignals act as peripheral proxies for cerebral activity during seizures. Different seizure types produce distinct physiological signatures across these signals, informing detection approaches. This relationship centers on two primary manifestations: motor activity and autonomic activation.

Convulsive seizures, such as generalized tonic-clonic events, produce clear motor signatures captured by accelerometry (ACC), which measures movement and posture changes characteristic of ictal events (Wang et al., 2025; Wu et al., 2024).

Simultaneously, seizures drive autonomic responses recorded through complementary biosignals. Electrodermal activity (EDA) measures sympathetic nervous system arousal via skin conductance changes (Jain et al., 2017; Miron et al., 2025). Cardiac activity provides particularly strong autonomic proxies: electrocardiography (ECG) captures heart rate and rhythm alterations (Ghaderi, 2025; Leal et al., 2017), while heart rate variability (HRV) quantifies autonomic balance dynamics (Mason et al., 2024; Pavei et al., 2017).

In wearable applications, photoplethysmography (PPG) serves as a practical alternative for cardiac monitoring, deriving pulse rate and variability though with greater motion sensitivity than ECG (Chen et al., 2022; Seth et al., 2023).

Consequently, detection strategies combine these biosignals according to target seizure types, with multimodal fusion providing the most robust approach for ambulatory monitoring.

2.4 Evaluation Frameworks: From Retrospective Analysis to Prospective Validity

The clinical validity of a seizure detection algorithm is determined by the rigor of its evaluation, formalized in the ILAE's phased framework (Phases 0–4) (Beniczky et al., 2021). Retrospective studies (Phases 0-2), while foundational, risk significant data leakage and optimistic bias if they use non-chronological data splits, allowing future information to inform training (Kalousios et al., 2024; Wong et al., 2023). Pseudo-prospective validation addresses this by enforcing chronological order, training only on past seizures to test on subsequent ones, providing a more realistic performance estimate (Kalousios et al., 2024; Lu et al., 2025). The clinical gold standard is a prospective trial (Phase 3), which requires a locked algorithm, a video-EEG reference standard, and testing on new, unseen data from multiple centers. The final step is in-field evaluation (Phase 4) in patients' homes, where metrics like the false alarm rate per hour (FAR/h) become decisive for practical usability, as demonstrated by long-term studies like Nightwatch (Miron et al., 2025; Shum & Friedman, 2021).

3 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by (Webster & Watson, 2002) and PRISMA principles (Tugwell & Tovey, 2021)). The methodological approach uses a multi-stage search and transparent extraction/synthesis procedures as outlined below.

1. **Search strategy (Multi-stage, 2015–2025):**
   - *Scoping:* Google Scholar for broad coverage and identifying candidate papers.
   - *Formal queries:* IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.
   - *Citation chasing:* Backward and forward citation tracking on key papers to ensure completeness.
2. **Inclusion / Exclusion criteria:**
   - *Inclusion:* Studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.
   - *Exclusion:* EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Extracted dimensions populate a concept matrix with the following fields:

1. Setting: clinical vs ambulatory.
2. Modality: ECG/HRV, PPG, EDA, ACC.
3. Task: detection vs prediction.
4. Model class and fusion strategy.
5. Study phase: retrospective, pseudo-prospective, prospective.
6. Evaluation granularity: sample- vs alarm-based.
7. Windowing specifics (window length, stride, overlap).
8. Metrics: sensitivity, FAR/h, AUC, f1-score, latency.
9. Personalization.
10. Interpretability and safety considerations.
11. Data characteristics and dataset identity.
12. Missing data handling (imputation, exclusion) and synthetic data use.
13. Inference requirements (latency, computational load, hardware).

Prioritization Criteria: Studies including splits preserving patient-specific data, FAR/h reporting and external or cross validation will be prioritized in the synthesis. A PRISMA-style flow diagram will summarize and visualize screening and inclusion.

4 Search Strategy

4.1 PICO-Based Search Methodology

The literature search was structured using the PICO framework (Patient/Problem, Intervention, Comparison, Outcome) to identify relevant studies on deep learning for non-EEG epilepsy detection. The following key terms were used in combination with Boolean operators (AND, OR, NOT):

1. **P (Patient/Problem):** Epileptic seizures in need of detection or prediction.
   - *Keywords:* 'epilepsy', 'seizure', 'ictal', 'preictal'
   - *MeSH Terms (PubMed):* Epilepsy, Se54izures
2. **I (Intervention):** The diagnostic method based on wearable biosignals and deep learning.
   - *Biosignal Keywords:* 'electrocardiogram', 'ECG', 'heart rate variability', 'HRV', 'photoplethysmography', 'PPG', 'electrodermal activity', 'EDA', 'accelerometer', 'ACC'
   - *Technology Keywords:* 'wearable', 'wearable device'
   - *Method Keywords:* 'deep learning', 'machine learning', 'convolutional neural network', 'CNN', 'long short-term memory', 'LSTM', 'neural network'
   - *MeSH Terms (PubMed):* Wearable Electronic Devices, Electrocardiography, Deep Learning
3. **C (Comparison):** The explicit exclusion of studies focused solely on electroencephalography (EEG) as the primary modality.
   - *Exclusion Keywords:* 'EEG', 'electroencephalography', 'iEEG', 'intracranial'
   - *Search Logic:* The `NOT` operator was applied to this group in all database searches.
4. **O (Outcome):** Metrics quantifying the performance of a detection or prediction system.
   - *Keywords:* 'detection', 'prediction', 'sensitivity', 'specificity', 'false alarm rate', 'FAR', 'area under the curve', 'AUC'

/subsectionExploration

Due to the insufficient amount of papers gathered from systematic review an exploration was done on the already existing papers. The exploration consisted of graph tools to visualize backlinks and gemini deep research feature to find relevant papers. Through this method an additional 4 papers were aquired bringing the total number of papers up to 13 and expanding the time frame to 2013-2026.

4.2 Practical Filters and Reproducibility

Consistent with the overall review scope, searches were restricted to **2015–2025** and **journal articles** with few exceptions made regarding **conference proceedings**. All database searches were documented (query string, platform, and execution date). Results were deduplicated prior to screening.

Note on exclusions: Because some non-EEG wearable studies rely on video-EEG for labeling, EEG-related terms were used to filter *EEG-only modalities*, not to exclude studies that mention EEG solely as a reference standard.

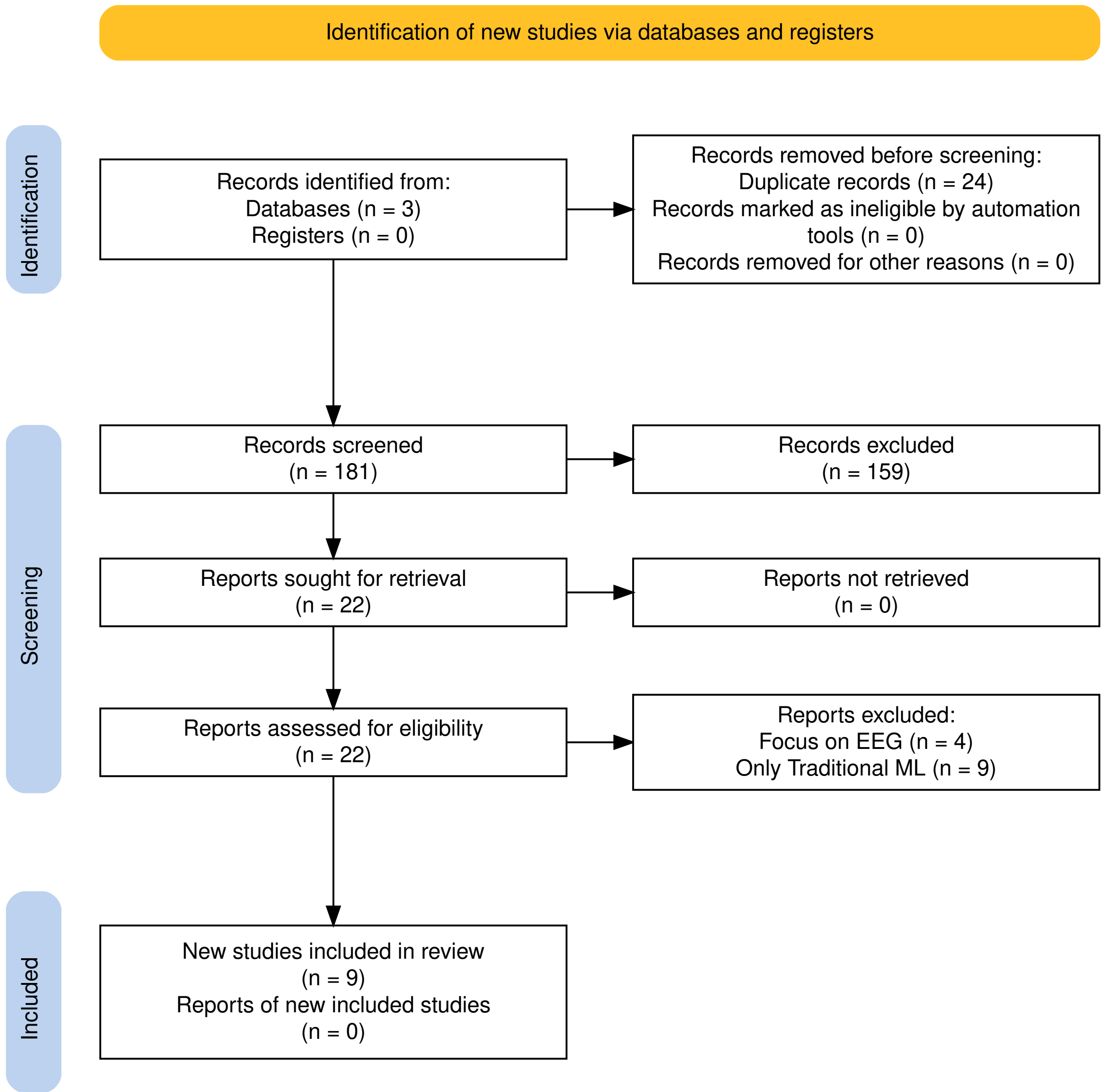

Figure 1: PRISMA flow diagram of the study selection process.

5 Study Comparison Following Webster and Watson

Following the concept matrix approach proposed by (Webster & Watson, 2002), this section presents a structured comparison of the 13 identified studies on wearable seizure detection and forecasting. The matrix organizes studies along key dimensions relevant to clinical translation and technical implementation.

5.1 Concept Matrix: Study Overview

Table 1 provides a comprehensive comparison of all included studies, organized by primary objective, methodology, and outcomes. The matrix enables cross-study analysis of trends in sensor modalities, algorithmic approaches, and validation strategies across the 2013–2026 period.

Table 1: Webster & Watson Concept Matrix: Wearable Seizure Detection and Forecasting Studies (2013–2026)

Study	Objective	Design	Sample	Device	Loc.	Algorithm	Sens	Spec	FPR	Key Limitation
Detection Studies										
Spahr et al. 2025	Convulsive detection multi-center	Prospective	n=384, 49 CSs	Empatica E4 (ACC)	Wrist	Ensemble 1D CNN	96%	–	<1/8d	EMU, offline
Reintjes et al. 2025	ECG anomaly detection	Retro	n=120, 856 szs	Single-lead ECG	Chest	Matrix Profile	98.16%	–	1.91/h	High FAR
Fine 2025	Tonic detection	Phase 1	n=15/3	6-axis band	Wrist	ANN (594 feat)	100%	–	0.23/n; 0.023/h	Small test
Dong et al. 2026	Nocturnal severe szs	Prospective	n=68, 1846 szs	NightWatch	Arm	CNN-LSTM	71.6%	–	0.165/h	Single-center
Wang et al. 2025	Sz detection (acc: 97.8%)	Retro	n=28, 62 szs	Biovital-P1	Wrist	LSTM+SVM/LDA	–	–	8.5/24h	Hospital only
Singh Rathore 2024	Multimodal	Single pt	1 pt, 6 hrs	EDA+ACC+HR	–	MLP, SVM, KNN	96.8%	–	–	Single patient
Elemam et al. 2025	Q+HRV+Audio	Cross-sect	Q:198, HRV:30	Camera PPG	Finger	CNN, rule based	95.1%	97.1%	–	Small samples
Borujeny 2013	Wireless ACC	3-pt	3 pts, 20 szs	MICAz ACC	Arm/thigh	KNN k=5	100%	–	0	Very small
Forecasting Studies										
Vieluf et al. 2025	Diary+wearable	Retro	n=70, 5437 d	Embrace	Wrist	DNN+harmonic	82%	67%	–	Daily res only
Meisel et al. 2020	LOSO forecast	LOSO CV	n=69, 452 szs	Empatica E4	Wrist	LSTM+1DConv	51.2%	–	TiW 43.7%	Pediatric
Stirling 2021	HR cycle forecast	Retro+pseudo	n=11, 136 szs	Fitbit	Wrist	LSTM+RF+LR	–	–	AUC 0.74	Self-report
Nasseri 2021	Ambulatory forecast	Retro	n=6	Empatica E4	Wrist	LSTM 4-layer	AUC 0.75	–	TiW 0.9–7.2h/d	RNS required
Ode et al. 2023	RRI prediction	Retro	n=66, 85 szs	ECG	–	Self-Att AE	74%	–	0.85/h	FPs in some pts

Note: ACC = accelerometer; ANN = artificial neural network; AE = autoencoder; CNN = convolutional neural network; CS = convulsive seizure; DNN = deep neural network; ECG = electrocardiogram; EDA = electrodermal activity; HR = heart rate; HRV = HR variability; LOSO = leave-one-subject-out; LSTM = long short-term memory; PPG = photoplethysmography; RRI = R-R interval; Sens = sensitivity; Spec = specificity; sz(s) = seizure(s); FPR = false positive rate; FA = false alarm; TiW = time in warning; – = not reported.

5.2 Dimensional Analysis

5.2.1 Sensor Modality Trends

- **Wrist-worn accelerometry** dominates (8/13 studies), leveraging commercial devices (Empatica E4, Fitbit, Embrace).
- **Cardiac signals** (ECG, PPG, HRV) show increasing adoption for preictal detection, with 6 studies incorporating autonomic measures.
- **Multi-modal fusion** (ACC + EDA + PPG) appears in 4 recent studies, reflecting recognition that single modalities have limited specificity.

5.2.2 Algorithmic Evolution

- **2013:** Traditional ML (KNN, ANN) with handcrafted features (Borujeny et al., 2013).
- **2**020–2021: Emergence of LSTM and ensemble methods (Meisel et al., 2020; Stirling et al., 2021).
- **2023–2026:** Deep learning dominance with CNNs, attention mechanisms, and anomaly detection approaches (Ode et al., 2023; Reintjes et al., 2025; Spahr et al., 2025).

5.2.3 Validation Rigor

- **Patient-level preservation:** Only 4 studies use LOSO or patient-independent splits.
- **Prospective validation:** Only 3 studies (Dong et al., 2022; Elemam et al., 2025; Spahr et al., 2025) include prospective phases.
- **Real-world testing:** Limited to 2 studies (Nasseri et al., 2021; Vieluf et al., 2025); most remain in EMU settings.

5.2.4 Performance Metrics Reporting

Table 2 summarizes the heterogeneity in performance metric reporting, complicating cross-study comparison. Sensitivity ranges from 51.2% (forecasting mean) to 100% (tonic detection), while FPR reporting varies from per-hour to per-night, with several studies omitting this critical metric entirely.

Table 2: Performance Metrics Reporting Across Studies

Metric	Detection (n=9)	Forecasting (n=4)	Reported By
Sensitivity/Recall	8/9	2/4	Reintjes, Fine, Spahr, Dong, Wang, Singh Rathore, Elemam, Borujeny, Meisel, Ode
Specificity	1/9	0/4	Elemam only
FPR/FA rate	7/9	2/4	Spahr, Reintjes, Fine, Dong, Wang, Ode, Meisel, Nasseri, Stirling
AUC-ROC	1/9	2/4	Reintjes, Nasseri, Stirling
Accuracy	2/9	0/4	Wang, Singh Rathore
Precision	0/9	1/4	Ode (reported as precision)

5.3 Synthesis and Gaps

The concept matrix reveals several patterns:

1. **Clinical translation gap:** Despite high sensitivities (>90% in several detection studies), FPR remains clinically unacceptable for standalone deployment.
2. **Personalization vs. generalization:** LOSO approaches achieve lower but more realistic performance (51.2%) compared to patient-specific tuning, raising questions about generalizability.
3. **Diary integration:** (Vieluf et al., 2025) demonstrates that hybrid approaches (wearable + clinical data) may outperform wearables alone, suggesting a direction for ambulatory systems.
4. **Metric heterogeneity:** Inconsistent reporting (sensitivity vs. accuracy vs. AUC) impedes meta-analysis and clinical benchmarking.

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